Le rôle de la cryoablation percutanée dans les métastases osseuses : contrôle local, debulking et antalgie

Titre original : The New Ice Age of Musculoskeletal Intervention: Role of Percutaneous Cryoablation in Bone and Soft Tissue Tumors

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1. Pertinence Clinique & Lien avec l'Agenda Interventionnel (Cochin)

Directement applicable aux cryoablations de métastases osseuses et pariétales (paroi thoracique, bassin, pied, trapèze) programmées en HDJ Radio B.

2. Synthèse Clinique en Français

Résumé exécutif : Revue complète sur la cryoablation percutanée des métastases osseuses à visée palliative (antalgique / debulking) ou curative (maladie oligométastatique). Analyse des résultats cliniques, du couplage avec les thérapies systémiques (anti-angiogéniques comme le lenvatinib, immunothérapie) et des techniques de consolidation associées.

Analyse détaillée : La cryoablation s'est imposée comme la technique de choix pour les métastases osseuses douloureuses ou oligométastatiques, notamment celles comportant une extension dans les parties molles. La visualisation directe du glaçon au scanner permet de traiter des volumes complexes tout en protégeant la peau, la plèvre ou les nerfs adjacents. L'article discute également la gestion péri-procédurale des thérapies ciblées anti-angiogéniques (tels que le lenvatinib ou le cabozantinib dans le cancer du rein métastatique) afin d'optimiser la cicatrisation cutanée et d'éviter les complications hémorragiques.

3. Protocole Technique & Paramètres Opératoires

Suspension temporaire concertée avec l'oncologue des ITK anti-angiogéniques (ex: Lenvatinib 48–72h avant/après selon demi-vie et risque cutané) • Double cycle de congélation 10 min • Consolidation par cimentoplastie ou vissage si atteinte d'une zone de contrainte mécanique.

4. Points Clés à Retenir pour la Pratique

- Soulagement antalgique significatif chez > 80 % des patients rebelles aux morphiniques ou après échec de la radiothérapie externe.
- Taux de contrôle local à 1 an de 85 à 92 % dans les indications oligométastatiques (marges d'ablation ≥ 5 mm).
- Dans les localisations distales (pied, main) ou sous anti-angiogéniques (Lenvatinib), vigilance accrue sur la protection cutanée.
- Pour les douleurs pariétales thoraciques oncologiques, la cryoablation associe destruction tumorale et cryoneurolyse intercostale.

5. Original PubMed Abstract (Texte Intégral de l'Abstract)

In the rapidly evolving field of interventional oncology, minimally invasive methods, including CT-guided cryoablation, play an increasingly important role in tumor treatment, notably in bone and soft tissue cancers. Cryoablation works using compressed gas-filled probes to freeze tumor cells to temperatures below -20 °C, exploiting the Joule-Thompson effect. This cooling causes cell destruction by forming intracellular ice crystals and disrupting blood flow through endothelial cell damage, leading to local ischemia and devascularization.

Coupling this with CT technology enables precise tumor targeting, preserving healthy surrounding tissues and decreasing postoperative complications. This review reports the most important literature on CT-guided cryoablation's application in musculoskeletal oncology, including sarcoma, bone metastases, and bone and soft tissue benign primary tumors, reporting on the success rate, recurrence rate, complications, and technical aspects to maximize success for cryoablation in the musculoskeletal system.

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6. Full-Text Sections (Europe PMC Open Access JATS XML)

1. Introduction

Interventional oncology has emerged as an increasingly crucial component of the multidisciplinary team, providing innovative and minimally invasive treatment approaches for various types of musculoskeletal tumors [1,2]. One such innovative technique, CT-guided cryoablation, is increasingly gaining recognition for its effectiveness, precision, and improved patient outcomes [3,4].

Cryoablation is a minimally invasive procedure that uses one or more probes filled with compressed gas (usually argon) to cool tumor tissue to very low temperatures [5]. This cooling is achieved by exploiting the Joule–Thompson effect, where rapid decompression of the gas surrounding the probe tip leads to temperatures below −20 °C [6,7]. The cooling damages the cells by forming intracellular ice crystals that destroy them, as well as impairing the blood supply by damaging the endothelial cells, which results in local ischemia and devascularization (Figure 1). Its integration with CT (Computed Tomography) technology allows for more accurate targeting of tumor cells, reducing collateral damage to surrounding healthy tissues, and minimizing postoperative complications [3,8,9].

Sequence of events showing the cryoablation technique. The cryoprobes are inserted inside the lesion (A–D) until it is all covered by ice (E). The cooling damages cells causing ischemia and devascularization, which result in lesion destruction and volume reduction (F–H).

This article aims to report on the use of CT-guided cryoablation in treating bone and soft tissue tumors, including sarcoma, metastases, and benign primary tumors, and the potential benefits and challenges associated with this novel therapeutic approach, including recent technical developments in cryoablation techniques. We will explore current research and clinical studies, highlighting the impact of this technique on cancer treatment and patient quality of life, and its promising role in the future of musculoskeletal oncology.

2. Malignant Bone and Soft Tissue Tumors

Percutaneous cryoablation is becoming an increasingly accepted option within the multidisciplinary sarcoma board for the treatment of primary bone and soft tissue tumors, applicable for selected cases. Despite surgical intervention being the mainstay for treating primary, non-metastatic bone, and soft tissue tumors, the local control of recurring bone and soft tissue sarcoma (STS) continues to be a challenging task. It mainly hinges on the disease prognosis as per the guidelines of the European Society for Medical Oncology (ESMO) [10,11]. Surgical resection is the common protocol for localized conditions, while chemotherapy or radiation therapy may be employed for more extensive diseases or recurrences [10,11,12]. Lately, minimally invasive techniques such as radiofrequency ablation, microwave ablation, or cryoablation have been proposed as potential surgical alternatives for some selected recurrent bone and soft tissue tumors [13,14,15,16,17]. Studies evaluating the role of cryoablation in the management of malignant bone and soft tissue tumors are summarized in Table 1.

Studies evaluating the role of cryoablation in the management of malignant bone and soft tissue tumors.

VAS: visual analogue scale; PFS: progression-free survival; OS: overall survival; DFS: disease free survival.

Some initial studies evaluated the therapeutic effect of Cryoablation for the treatment of a variety of primary bone and soft tissue malignancies with promising results; however, the scientific evidence is still limited. Moreover, there is a recognized need for the standardization of selection criteria for percutaneous cryoablation. Lippa et al. [12] aimed to identify these criteria, finding high agreement for all proposed criteria between two readers. Eligibility for cryoablation was significantly associated with tumors located deeply, with great axes ≤ 5

cm, high local tumor aggressiveness, and a diagnosis of differentiated myxoid liposarcoma or myxofibrosarcoma.

Some retrospective studies have reported on the effectiveness and safety of percutaneous cryoablation in the treatment of recurring retroperitoneal soft tissue sarcomas (RPSs). RPSs, which make up approximately 0.15% of all cancers, originate in the retroperitoneum but not from its main organs. Their proximity to critical structures makes them challenging to manage. Surgery is the primary therapeutic approach for localized cases, leading to a survival rate of about 60% over five years [21,23]. Nonetheless, complications with the removal of RPSs can impact survival and lead to recurrence. For recurrent cases, re-operation is recommended, but it is more difficult, and the additional benefits of chemotherapy and radiotherapy are debatable. In some selected cases, cryoablation has proven safe and effective as a palliative treatment for RPSs and could be included in the armamentarium of the sarcoma board. Fan et al.'s retrospective analysis of data from 72 patients [21] primarily noted grade 1 and 2 adverse events, with fever being the most common, particularly in the group with larger tumors. The study found a median progression-free survival (PFS) and overall survival (OS) of 37.0 ± 7.7 months and 43.0 ± 5.9 months, respectively. A significant association was observed between tumor size and both PFS and OS, with smaller tumors linked to longer survival. No significant variance was seen in response rates between small and large tumor groups. Another study by Fan et al. found that percutaneous cryoablation significantly alleviated local pain in patients with recurring RPSs, with immediate relief more common in the group with smaller tumors [22].

Kurup et al. [20] documented the use of cryoablation in treating recurrent sacrococcygeal tumors, with the findings suggesting that this method was secure and relatively effective for local management or pain reduction (Figure 2). Similarly, Susa et al. [18] assessed the clinical outcomes of CT-guided cryoablation for recurring or metastatic malignant bone and soft tissue tumors, involving nine patients over a median observation period of 24.1 months. Although they reported promising outcomes, the study's effectiveness was significantly limited by the small patient group. Additionally, Li et al. [19] recorded favorable results following the application of CT-guided argon–helium cryoablation in the treatment of sacral chordoma. All the patients achieved a complete response (CR), and there was a considerable improvement in chordoma-related symptoms post-treatment. The average score on the visual analog scale improved from 7.3 before treatment to 4.2 post-treatment, and the median progression-free survival was 36.8 months.

A 55-year-old man, cryoablation in a recurrent chordoma showed in an axial T2 MR image ((A)—arrowheads) and axial postcontrast CT image ((B)—arrowheads), for local tumor control. The cryoprobe was placed into the lesion under CT (C) and US guidance, with subcutaneous hydro dissection performed to protect the skin. The ice ball encompassed the entire lesion (D). Also, note the aspect of the ice ball on ultrasound ((E)—arrowheads).

3. Bone Metastases

Bone frequently becomes a site of metastases, ranking as the third most common area for metastatic carcinoma after the lungs and liver [24]. Given the typically low survival rates of patients, therapeutic options are usually limited, and surgical resection is rare [24,25]. However, skeletal complications such as severe and chronic pain, spinal cord compression, and pathological fractures can greatly impair a patient's quality of life [26,27,28,29,30,31,32]. The escalating interest in thermal ablation techniques for handling bone metastases has positioned cryoablation as a top-tier approach. This is due to its capacity to treat extensive and irregularly shaped pathological tissues in real time, while causing less pain compared to heat-based ablative methods like radiofrequency and microwaves [33,34,35]. The current body of literature mostly explores the palliative effect and local tumor control of cryoablation for bone metastases, which makes this technique a useful tool in the multidisciplinary management of cancer patients.

Studies assessing the role of cryoablation in managing bone metastasis are outlined in Table 2.

Studies evaluating the role of cryoablation in the management of bone metastases.

* PFS: progression free survival; VAS: visual analogue scale; NRS: numerical rating scale; LTC: local tumor control; DFS: disease-free survival; OS: overall survival; RFA: radiofrequency ablation; CA: cryoablation; BM: bone metastases; LPFS: local progression free-survival; RT: radiotherapy; CT: computed tomography.

The clinical effect in pain reduction and safety of cryoablation for metastatic bone disease has been investigated by several studies in recent years [40,43,44,46,48] (Figure 3). A recent multicenter prospective study by Jennings et al. [38] assessed the clinical efficacy of cryoablation as a pain palliating method for patients with metastatic bone disease. The main goal was the pain score change from pre-treatment to the eighth-week follow-up, with participants monitored for 24 weeks post-treatment. A cohort of 66 participants (average age 60.8 years, 53% male) was recruited and underwent percutaneous cryoablation; 65 completed the follow-up. The average change in pain score from baseline to the eighth week decreased by 2.61 points. Average pain scores improved by 2 points at the first week and attained clinically significant levels (a decrease of more than 2 points) post the eighth week, with scores continuing to improve throughout the follow-up period. Quality of life was enhanced, opioid doses were steady, and functional status remained unchanged over six months. Severe adverse events were reported in three participants. Overall, the study suggested that cryoablation of metastatic bone tumors provided quick and long-lasting pain relief, enhanced quality of life, and presented an alternative to opioids for managing pain. The procedure was generally well received, and the severe adverse events were not directly linked to the procedure.

Axial T2 MR image of a 51-year-old woman with metastases from leiomyossarcoma (A) in the pelvis treated with cryoablation (C) for palliative intent. 18F-FDG PET/CT scan performed before the procedure demonstrates the pathologic radiotracer uptake in areas of viable tumor (B). The procedure was performed in an attempt to cover these areas and the post-procedure ice balls are visible as hypodense circles (arrowheads—(D,E)).

In a similar way, Arrigoni et al. [37] reviewed the results of 28 cryoablation procedures conducted over a span of seven years in their radiology department. These procedures included 17 palliative treatments and 11 curative treatments. A follow-up study after three months showed a substantial decrease in pain, as indicated by the average Visual Analog Scale (VAS) scores declining from 6.9 to 3.5. In patients treated for local tumor control, the study showed either stability or shrinkage of the lesion in 10 out of 11 patients, with no significant complications recorded.

In a prospective trial analysis, Callstrom et al. [52] evaluated the safety and effectiveness of cryoablation in pain reduction, enhancement of daily activities, and decreased usage of analgesics in patients with painful metastatic bone lesions. All eight patients who were on narcotic medication before the procedure reported a decrease in these medications after cryoablation, with no serious complications noted.

4. Benign Bone Tumors

Osteoid osteoma (OO) is a small, benign tumor primarily found in the bones of young people and children. Even though it only accounts for approximately 10% to 12% of all benign bone tumors, it can significantly affect the quality of life, causing pain and bone deformity, especially in children [65,66,67]. The treatment for OO has seen a considerable evolution over the years. Traditional surgical removal was once the main treatment approach, but technological advancements have facilitated a shift toward less invasive methods like radiofrequency ablation (RFA) [68,69,70,71].

Meng et al. [72] conducted a study comparing the safety and effectiveness of percutaneous CT-guided cryoablation of OO to surgical curettage. Both treatment approaches reported a 100% technical success rate. However, patients treated with cryoablation spent significantly less time in the hospital than those undergoing surgery, and both groups showed notable improvement in postoperative Visual Analog Scale (VAS) pain scores.

Percutaneous cryoablation for OO treatment is increasingly being acknowledged as a safe and effective technique (Figure 4). Research assessing the role of cryoablation in managing osteoid osteoma is summarized in Table 3.

Cryoablation of osteoid osteoma of the elbow in a 28-year-old man: Coronal CT image shows the nidus (arrowheads) with surrounding sclerosis (A). Axial CT images acquired during the procedure show the cryoprobe placed inside the lesion (B) and the low-attenuation ice ball encompassing the lesion (C). Sagittal T1 MRI at 1-month follow-up shows signal change corresponding to the ablation area (arrowheads—(D)). Patient reported considerable improvement in pain.

Two studies, one by Coupal et al. [73] and the other by Santiago et al. [74], were among the first to report the use of CT-guided percutaneous cryoablation in treating OO in adults. Coupal et al. reported a 100% clinical and technical success rate, with the average numeric pain scores decreasing significantly from 7.4 before the procedure to 0.3 at the follow-up. Similarly, Santiago et al. reported a substantial reduction in median VAS scores from 8 pre-procedure to 0 at the primary and secondary follow-ups. Both studies reported minor complications only, emphasizing the safety of the procedure.

The safety and efficacy of this procedure also extend to the pediatric and adolescent population, as shown in the study by Whitmore et al. [75]. It was found that immediate, short-term, and long-term clinical success was achieved in 96.4%, 96%, and 90.5% of patients, respectively. The median pain score dropped from 10 to 0 after the procedure. This study again highlighted that the procedure was technically feasible and had no major complications.

Miyazaki et al. [76] took a step further by conducting a prospective trial to assess the safety of percutaneous cryoablation for OO. They found no major complications associated with the procedure, while minor adverse events were observed in 22–67% of patients. The average Numeric Rating Scale (NRS) score significantly reduced from 7 before the procedure to 0 a year after treatment. These results suggest that cryoablation can be a safe and effective treatment for OO.

Percutaneous cryoablation has some key advantages over traditional OO ablation techniques like radiofrequency ablation. For instance, it allows the lesion to be treated by positioning the cryoprobe next to the bone cortex instead of penetrating the nidus. Additionally, cryoablation for OO can often be performed under local anesthesia and sedation because it tends to be less painful and more tolerable than radiofrequency or microwave ablation. These advantages were highlighted in a study by Le Coroller et al. [77] where cryoablation was carried out under local anesthesia and conscious sedation for 60% of the patients (30 out of 50). The cryoprobe was positioned outside the bone to cover the lesion, thus avoiding direct penetration of the nidus. The overall clinical success rate was 96% (48 out of 50 patients). Of the two patients who did not achieve clinical success, one had incomplete pain relief, and the other experienced a recurrence of the osteoid osteoma at 11 months. The latter was successfully treated with a second cryoablation procedure.

5. Desmoid Tumors

Desmoid tumors are rare benign tumors originating from musculoaponeurotic structures [85]. Despite their benign nature, they can display local aggressiveness, causing disability and sometimes pain. The ESMO advises initial observation and subsequent medical treatment for progressing tumors. Cryoablation, an interventional radiology technique, is recommended for desmoid tumor patients due to its ability to induce cell death through multiple cycles of freezing [86,87,88] (Figure 5).

A 46-year-old patient with desmoid tumor of the abdominal wall showed in axial T2 fat sat MR image ((D)—arrowhead). Axial (A) and sagittal oblique (B) CT images showing the cryoprobe inside the lesion and the post-procedure ice ball containing the whole lesion (C). Axial T2 fat sat MR image 3 months after treatment shows an increase in T2 signal inside the lesion, suggesting necrosis (E), and the control with 6 months shows no residual tumor (F).

Given the high recurrence rate, surgery's role in treating desmoid tumors is restricted only to selected cases [89]. However, percutaneous cryoablation has recently come into the spotlight as a promising treatment option, demonstrating positive results in safety, effectiveness, and symptom alleviation [90,91,92]. A systematic review and meta-analysis conducted by Vora et al. [93] scrutinized the safety and effectiveness of cryotherapy in treating extra-abdominal desmoid tumors. The findings suggested that the combined estimated percentage of

major and minor complications was 4.2% and 10.2%, respectively, signifying a relatively low risk. The rate of disease non-progression across all the studies was significantly high at 85.8%. The one-year and three-year progression-free survival rates stood at 84.5% and 78.0%, respectively. Regarding pain management, the study discovered that 87.5% of patients who initially had a Visual Analog Scale (VAS) score of three or more experienced a reduction in their VAS score by at least three points. Furthermore, between 37.5% and 96.9% of patients reported experiencing partial or complete relief of symptoms. The study definitively concluded that cryotherapy is a safe and effective treatment for extra-abdominal desmoid tumors, providing short to medium-term effectiveness, comparable to traditional treatments. Moreover, another systematic review by Cazzato et al. [5] emphasized the benefits of considering cryoablation as a first-line treatment, suggesting it could potentially enhance clinical outcomes. A summary of studies exploring the role of cryoablation in managing desmoid tumors is provided in Table 4.

Studies evaluating the role of cryoablation in the management of desmoid tumors.

PFS: progression-free survival; DFS: disease-free survival; ET-V: enhanced tumor volume; VAS: visual analogue scale.

Mandel et al. [98] conducted a study aimed at comparing the outcomes of patients with primary and recurrent extra-abdominal desmoid tumors undergoing percutaneous cryoablation with those undergoing surgical treatment. In this study, 22 patients treated with cryoablation were compared with 33 surgical patients. The median monitoring periods were 16.3 months for cryoablation and 14.9 months for surgery. Local recurrence-free survival (LRFS) rates for two years stood at 59% for cryoablation and 71% for surgery, although the median LRFS for surgery was not met. Two-year disease control reached 85% for all patients, with no significant difference between the cryoablation and surgical groups. Notably, the study revealed that no local recurrences happened after the first cryoablation in patients who had none or one of the following risk factors: tumor size larger than 5 cm, age less or equal to 25 years, or a history of locally recurrent disease. In contrast, all patients presenting with two or more of these risk factors experienced local recurrence after the first cryoablation. The study's main conclusion is that percutaneous cryoablation of primary and locally recurrent extra-abdominal desmoid tumors provides a satisfactory progression-free survival and long-term disease control, comparable to that of surgical intervention.

Another study by Yan et al. [88] assessed the safety and effectiveness of cryoablation in treating desmoid tumors over a decade. The results demonstrated that cryoablation was successful in shrinking the volume of the viable tumor, with a median change of –43.7%. The majority of patients exhibited a partial response (61.5%) or stable disease (30.8%). Symptomatic relief was attained in 96.9% of cases, and there was a low occurrence of major complications (2.4%). The authors concluded that cryoablation seems to be a safe and effective treatment for local control of extra-abdominal desmoid tumors.

Although technically benign tumors, desmoid tumors have a high potential for local recurrence and local invasion. For this reason, they should be treated as malignant lesions, and to obtain a curative effect, all the lesions should be covered by the ice ball, leaving some margins if possible. In this regard, Schmitz et al. [95] shared their extensive 10-year experience using percutaneous cryoablation for treating desmoid tumors. They reported no serious complications and significant shrinkage of 95.7% of the tumors treated. However, in all instances of residual or advancing disease, recurrence happened at the edge of the previously treated tumor.

6. Technical Consideration

Cryoablation planning requires pre-procedural imaging, ideally within a month prior to the procedure. Thin-section CT is recommended for bone lesions, while MRI with dynamic contrast enhancement and high spatial resolution techniques offer higher sensitivity and specificity for soft tissue lesions [100].

When the lesion is not deeply located and is outside the bone, as in soft tissue tumors, the ultrasound-guided placement of the needles may expedite the process and reduce patient exposure to ionizing radiation [100,101,102]. Hydrodissection is a valuable technique, used to create a safe margin between the tumor and adjacent critical structures. To enhance visibility during control CT scans, it is recommended to use iodinated

water for hydro dissection. This allows the operator to clearly delineate the dissected area [58].

During the ablation phase, real-time visualization of the ice ball using ultrasound can assist in limiting the number of CT scans. However, the complete ice ball cannot be fully visualized due to shadowing; therefore, visualization through CT is essential [103]. When performing cryoablation near neural structures, such as the spine or peripheral nerves, continuous monitoring of evoked potentials is necessary. If a decrease in potentials is observed, the procedure should be halted immediately, as the ice ball may continue to grow for several minutes after the cycle's interruption. This strategy aids in preventing neural damage, a significant potential complication of cryoablation [59].

7. Discussion

The current body of literature extensively examines the application of percutaneous cryoablation for various types of malignant and benign bone and soft tissue tumors. The flexibility and precision of this treatment method, combined with its relative safety, make it a promising option for a variety of tumor types and sizes.

Regarding treatment effectiveness and safety, the visualization of the ice ball offers high precision. It permits the simultaneous use of many cryoprobes, allowing for the ablation of large lesions (>5 cm), the creation of ice balls with a diameter greater than 8 cm, and the shaping of the ablation zone to the shape of the target lesion through varied geometry of probe placement. In general, when multiple probes are used, they are placed approximately 2 cm apart within the tumor and 1 cm from the outer tumor margin. It is also less painful than RFA during ablation and after treatment, with a shorter hospital stay [104,105].

For primary bone and soft tissue tumors, the work by Lippa et al. [12] underscores the necessity of standardized selection criteria for eligible patients. Their study suggests that deep tumors with high local aggressiveness and a diameter ≤5 cm may be best suited for cryoablation, particularly in cases of differentiated myxoid liposarcoma or myxofibrosarcoma.

Cryoablation also shows considerable potential in managing recurrent retroperitoneal soft tissue sarcomas (RPSs). As observed by Fan et al. [21], adverse events associated with the procedure are predominantly minor, with fever being the most common. This safety profile positions cryoablation as a feasible alternative to surgery, which can often present more significant challenges and complications.

When it comes to the treatment of recurrent sacrococcygeal tumors and chordoma, research by Kurup et al. [20], Susa et al. [18], and Li et al. [19] points to the relative effectiveness and safety of cryoablation for local control or pain relief. However, the limited sample size in Susa et al.'s research [18] indicates a need for further study with more extensive participant groups.

Percutaneous cryoablation has also shown potential in the management of bone metastases. Even with typically low survival rates and complex treatment scenarios for these patients, cryoablation appears to be a beneficial approach for pain alleviation, local tumor control, and possibly even curative intent in certain situations [33,34,35,38,49,52,55]. However, Cazzato et al.'s [49] findings suggest that the best results might be achieved with bone metastases smaller than 2 cm.

Cryoablation also has a role in treating benign bone tumors. With respect to osteoid osteoma, studies by Meng et al. [72], Coupal et al. [73], and Santiago et al. [73] confirm the safety and efficiency of percutaneous cryoablation compared to traditional surgical procedures. Regarding osteoblastoma, the work by Cazzato et al. [78] underscores the importance of comprehensive safety precautions due to the common closeness of vital structures.

In treating symptomatic bone cysts and aneurysmal bone cysts, studies by Alkuhaimi et al. [83] and Arleo et al. [84] depict the technical success and beneficial outcomes of cryoablation. Still, Arleo et al.'s findings suggest fewer procedures might be required with cryoablation compared to other treatments, such as doxycycline sclerotherapy.

8. Conclusions

Percutaneous cryoablation is a reliable and successful method for treating a range of bone and soft tissue tumors, being increasingly incorporated into the multidisciplinary decision-making process of tumor boards. The procedure’s benefits encompass prompt alleviation of pain, enhancement of life quality, and limited unwanted side effects. Precision in needle placement and ice ball visualization, facilitated by imaging guidance, further enhances the safety and accuracy of the procedure, making it a viable option even for lesions located near critical structures. Despite these benefits, the decision to perform cryoablation should consider tumor location, size, and aggressiveness.

Cryoablation has demonstrated substantial potential in handling a variety of musculoskeletal tumors, from retroperitoneal soft tissue sarcomas to sacral chordomas, bone metastases, and benign bone and soft tissue tumors, underlining its adaptability. However, more research is required to fully understand the extent of cryoablation’s capabilities. Future research directions should focus on establishing standardized patient selection criteria, examining the possibilities when cryoablation is paired with other localized or systemic treatments, and implementing more long-term studies and randomized controlled trials to evaluate its enduring effectiveness and safety.